HX-200 Centrifugal Pump Operation and Maintenance Manual

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1. Scope

This manual covers installation, operation and maintenance of the Model HX-200 horizontal centrifugal pump, installed as asset Pump P-204 in the Process Unit 2 cooling-water loop.

This revision supersedes Revision B. Changes: maximum process temperature raised to 120 deg C following seal upgrade per ECN-0412; seal face material changed; grease interval shortened.

2. Design and operating data

Parameter	Value	Unit
Rated flow	120	m3/h
Rated head	45	m
Maximum process temperature	120	deg C
Maximum casing pressure	10	bar
Mechanical seal material (faces)	Silicon Carbide vs Silicon Carbide	-
Bearing type	Deep groove ball, 6310	-
Bearing grease replacement interval	3000	operating hours

3. Operating limits

The pump shall not be operated with process fluid above 120 deg C. Continuous operation above this limit risks seal face damage and loss of seal integrity. Vibration velocity at the bearing housing must remain below 4.5 mm/s RMS.

4. Startup procedure

1. Verify suction valve fully open. 2. Prime the casing. 3. Check shaft rotation by hand. 4. Start the motor with discharge valve 10 percent open. 5. Open discharge valve slowly while monitoring motor current.

5. Maintenance

Replace bearing grease every 3000 operating hours. Inspect the mechanical seal at every planned shutdown. Replace seal faces if leakage exceeds 5 drops per minute.

6. Troubleshooting

Symptom	Likely cause	Action
Seal leakage	Face wear or overtemperature	Inspect faces, check process temperature log
High vibration	Bearing wear or misalignment	Check alignment, replace bearing
Low flow	Impeller wear or air ingress	Inspect impeller, check suction line